

Day 12 (18/11/2025)

§ Nested while loop - one while loop runs inside another

Syntax of nested while loop -

Initialization of outer while

while (cond) : # outer while

initialization of inner while

while (cond) : # inner while

Statements of inner while

inc/dec of inner while

Statements of outer while

inc/dec of outer while

1 to 5 Table.

i = 1

while (i <= 10) :

j = 1

while (j <= 5) :

print(f"{j} X {i} = {i*j}, end = '\t')

j = j + 1

print()

i = i + 1

$$1 \times 1 = 1$$

$$1 \times 2 = 2$$

$$1 \times 3 = 3$$

⋮

$$1 \times 10 = 10$$

$$2 \times 1 = 2$$

$$2 \times 2 = 4$$

$$2 \times 3 = 6$$

⋮

$$2 \times 10 = 20$$

$$3 \times 1 = 3$$

$$3 \times 2 = 6$$

$$3 \times 3 = 9$$

⋮

$$3 \times 10 = 30$$

$$4 \times 1 = 4$$

$$4 \times 2 = 8$$

$$4 \times 3 = 9$$

⋮

$$4 \times 10 = 40$$

$$5 \times 1 = 5$$

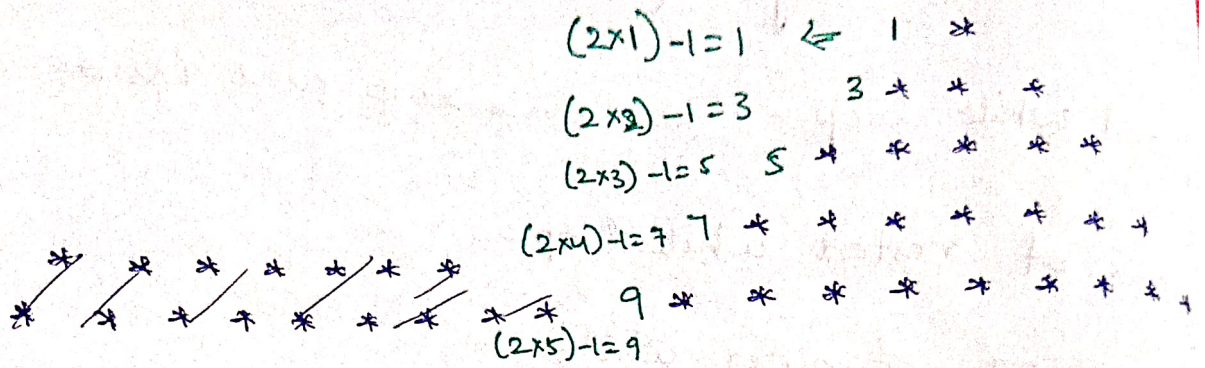
$$5 \times 2 = 10$$

$$5 \times 3 = 15$$

⋮

$$5 \times 10 = 50$$

Q)



rows = 5

i = 1

while (i <= rows):

j = rows - i

while (j > 0):

print(" ", end=" ")

j = j - 1

k = 1

while k <= (2 * i - 1):

print("*", end=" ")

k = k + 1

print()

i = i + 1

Syntax of while loop inside a for loop -

for i in (start value, stop value, inc/dec):

initialization

while condition:

statements of while loop

inc/dec while

statements of for

Syntax of for loop inside a while loop.

initialization

while(cond):

for var in range():

Statements of for

Statements of while

in/dec of while.